

AI-Powered Automated Evaluation of Handwritten Student Answers Using OCR and BERT-Based Semantic Similarity

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Abstract— Marking of handwritten answer sheet is also time consuming, inconsistent and subject to human bias. The AI Answer Sheet Evaluator solves these issues by presenting a new model of automated grading with Artificial Intelligence (AI) and Natural Language Processing (NLP). The system uses OCR programs like Tesseract and TrOCR to turn the handwritten student answers into a digital text. Semantic similarity models such as Sentence-BERT and GPT are then used to evaluate extracted text by comparing answers of students with answers given by the instructor using the models. Judging is done according to accuracy, relevancy, completeness and coverage of important points. This model is unlike traditional systems that only provide binary scoring since it supports partial marking whereby fair distribution of mark can be made to partly correct answers. With OCR and NLP, the system is effective in handling big amounts of scripts in addition to being transparent and objective. This smart and scalable solution will help bring about a transformation in the field of academic assessment in schools, universities, and extensive test-taking facilities.

Keywords—OCR, BERT, NLP, Automated Answer Evaluation, Semantic Similarity, Deep Learning, Answer Script Processing, Educational Technology, Image-to-Text Conversion, Academic Assessment

I. INTRODUCTION

This template, modified in MS Word 2007 and saved as a Answer sheets done by hand are human based and time consuming programs that are difficult to implement in learning institutions. The manual assessment is highly susceptible to errors, variations and subjective bias particularly with a large number of scripts. As the number of students and examinations increases, it has been harder to ensure that there is accuracy and fairness in grading. These problems present the necessity of a computerized system which is capable of delivering uniform, quick and objective assessment. To meet this requirement; the AI Answer Sheet Evaluator applies the concept of Optical Character Recognition (OCR) and Natural Language Processing (NLP) to a set of grading procedures to automate the scoring. With the help of the system, OCR technologies Tesseract and TrOCR are used to translate handwritten responses into a

digital form, where they can be further analysed. The system compares the responses of students semantically with teacher- provided model answers using advanced models of NLP (Sentence-BERT and GPT). With this project, the partial marking mechanism is presented, as opposed to traditional automated systems of grading which depend on the possible matching of keywords or merely awarding binary marks. It marks the responses on the basis of their correctness, relevance, completeness and key points with proportional marks awarded to partially correct responses. This will guarantee more balanced and subtle assessment and transparency. Moreover, it can be scaled and handle huge amount of hand written replies efficiently, thus it can be used in schools, universities and mass examinations.

II. LITERATURE SURVEY

In this paper, a survey can be viewed as a description of a handwritten text recognition system in detail, and the development of the techniques towards the incorporation of the handcrafted features, followed by the development of the techniques that evolved to the neural systems and transformer-based systems. It also discusses data preparation, model structures, learning methods, benchmarking patterns and unresolved issues such as limited data, document structure complexity, cross-language generalization that gives a structured information guide to the subsequent researches [1]. The paper proposes a proof-of-concept OCR system, which will transform models to identify text-process lawmaking petition handwritten signatures. It reports system design, signature text extraction and verification process and demonstrates how attention based architectures, as compared to existing OCR methods, excel in administrative and legal document processing in recognition accuracy, recognition robustness and recognition scalability. [2]. The method that is offered in this paper is an offline handwritten Chinese text recognition approach that is founded on purely convolutional neural networks and enhanced by residual and attention units. All the complex character structures are represented by the model, the information loss is reduced in the deep layers and the recognition accuracy is enhanced. Examinations show that there is an effective handling of massive-vocabulary Chinese characters to enhance solidness and efficiency [3].

In this paper, the term offline handwritten text recognition was used in reference to historical documents that contain some annotations. It suggests the application of transfer

learning, data augmentation, and semi-supervised training in order to improve the performance when the number of labeled text lines is extremely small and demonstrate that it can lead to higher accuracy and reduced labor is necessary in the process of heritage documents analysis. In this paper, the author proposes an improved GPT model, which is enhanced with OCR that may be used in the industrial environment to be able to extract text with a high degree of certainty [4]. It is based on a large language model with conventional OCR to fix recognition errors, noisy backgrounds and reading between the lines, leading to higher accuracy and robustness on complex industrial images and documents [5].

In this paper, SHIFA, an Arabic answer system on healthcare queries achieved in the context of SBERT embedding, was presented. It targets the retrieval of correct medical information through creation of semantic similarity between questions and answer. The paper will discuss data set creation, model training, and evaluation that is more precise and pertinent than the traditional way of creating data based on keywords and the baseline neural models. [6]. In this paper, the author offers an automated grader of short answers in Japanese in questions on the basis of BERT contextual embeddings. It compares semantically between student answers and reference answers, and it copes with the linguistic variation that is widespread among the learners. The model promotes the grading consistency and the fairness as well as has been shown to be effective to the Japanese language education as well as scalability of grading systems. [7]. This article provides a BERT-based aspect based sentiment analysis system on Arabic open-ended survey data. It defines the objects of opinion and the associated sentiments in the answers and takes care of the language complexity and dialect difference. In the case study, it shows that the application of traditional methods proves to be more precise and provides an extra insight that can be used to justify the decision making and determine the policy [8].

The given paper suggests a clever structure of descriptive responses analysis using machine learning and NLP. It provides comparatively consistent and objective scoring as compared to model answers by the preprocessing, feature extraction and semantic similarity analysis. It will solve such problems as the disparity in the sentence structure and synonyms, the research will assist in refining automated assessment in education and making it more efficient. [9]. In this paper, the automated solution of reviewing the answers is developed as an automated answer script system relying on OCR, deep learning and NLP. OCR makes the handwritten data digital, and the analysis of handwritten data grammar and semantics is done in NLP processes.

Deep learning enhances contextual knowledge, and, therefore, it is possible to score properly. The system reduces the manual grading, increases the consistency and facilitates the scalable and efficient grading in learning institutions [10]. The flexible program presented in this paper allows analyzing digitized answer scripts and that is, breaking down the scripts into questions rather than analyzing the whole script. The system that is based on image processing to retrieve specific response is more precise, transparent and consistent and able to handle partial response and various forms and therefore more efficient and reliable in digital educational assessment systems [11]. The present paper proposes a framework of scanned answer scripts assessment

based on a deep learning algorithm. They can be used even when the phrasing is changed or incomplete responses, after the semantic meaning and contextual accuracy is assessed by neural networks, after the digitalization and preprocessing process. The approach reduces the level of manual labor, reduces bias, and gives a massive and reliable solution to academic assessment [12].

The paper presents a method of NLP-based assessment of student responses to semantically and syntactically analyze the answers. The relevance, partial accuracy and contextual meaning are determined through preprocessing, word embeddings and similarity measures. The strategy reduces hand scoring, enhances objectivity and enables evaluation of descriptive responses in education to be done at scale and reliably [13]. The paper describes the idea of automated analysis of descriptive responses with the methods of natural language processing. It compares student responses with model responses through the methods of semantic similarity, key words and embeddings extraction of the answer. This study demonstrates high objectivity, scalability and consistency in scoring as compared to hand scoring [14]. In this paper, I will discuss how to assess subjective responses using machine learning and natural language processing (automation). It merges feature extraction, semantic analysis and predictive modeling to assess the responses of students with a view of enhancing the grades reliability, reducing the manual effort, and providing scalable and objective grading to educational evaluation [15].

III. PROPOSED SYSTEM

The suggested approach introduces an automated AI-based evaluation system and assesses handwritten answer sheets on the basis of the possibilities of a combination of Optical Character Recognition (OCR), Natural Language Processing (NLP), and semantic similarity models. It involves the process of scoring, and students upload scanned image of their handwritten answers which go through a lot of preprocessing steps which include; grayscale, noise, skew and thresholding which helps to increase readability and improve the accuracy of OCR. The images are then processed with the created and refined image using sophisticated OCR engines such as Tesseract and TrOCR to precisely extract the text even with the differences in writing styles, spacing, or potential stroke patterns. After the text is extracted, it is cleansed and normalized by applying NLP preprocessing tasks such as tokenization, eliminating stop-words, stemming /lemmatization and punctuation-filtering. This is in order to make the content structured and to be semantically interpreted.

Sentence-BERT is used in the system to create quality sentence embeddings of the student question and the model answer provided by the instructor to allow deep semantic contrast as opposed to simple keyword matching. GPT also complements the assessment by being able to read between the lines, extract partially applicable knowledge, and extract obscure conceptual significance in paraphrased or partial responses. Embeddings are compared to determine the semantic similarity between two entities which is cosine similarity and a partial marking algorithm turns this score into proportional and fair marks. The final outputs, extracted text, semantic score and received marks are stored in the database and presented in separate teacher and student dashboards, which would guarantee transparency,

consistency, scalability, and a considerable decrease in the amount of human resources needed to conduct academic evaluation.

IV. SYSTEM DESIGN

A. Dataset Description

A sample size of 2,000 hand written answer sheets was created, and the sample involved science, social studies, and computer application subjects. In the samples, there is the handwritten answer of the student, as well as a reference answer provided by the teacher. The data were gathered among students who are studying in the 9th up to the 12th grades and the sample was diversified in terms of handwriting style, use of vocabulary and length of answers. In order to further reinforce the OCR part, more samples of handwriting were added together with the public data, such as the IAM Handwriting Database and NIST Special Database 19, which have around 20,000 word-level images. The additional datasets only provided supplementary data to perform OCR fine-tuning, and the custom academic dataset was applied, as the main source to evaluate the semantics. A panel of five seasoned educators scored the Ground-truth manually and, according to a standardized rubric, gave scores as ground-truth. Based on relevance and completeness the answer of each student was classified as either fully correct, partially correct, or incorrect. This rich and highly annotated collection of data allowed the high OCR accuracy and the effective semantic assessment in the process of training and testing of the models.

B. System Architecture

The The AI Answer Sheet Evaluator will have the design of an end-to-end automated pipeline, which will combine several modules to handle and analyze handwritten answers effectively. The system starts with the Input Module whereby the handwritten answer sheets of students are scanned or photographed and uploaded to the system. Preprocessing in the images is done to eliminate noise, improve contrast, and skew correction to improve OCR reading. The OCR Module makes use of complicated tools like the Tesseract and TrOCR to scan the text on the handwritten sheets and turn it into machine readable digital file as well as addressing the variety of handwritings. The Preprocessing Module subsequently purges the extracted text, through tokenization, lemmatization, stop word removal as well as OCR errors correction. This preconditions semantic analysis of the text.

The NLP and Semantic analysis module involves the use of models such as Sentence-BERT and GPT to come up with the embedding of both the student answers as well as the reference answers provided by the teacher. At the same time, Cosine similarity is calculated to estimate the semantic proximity between responses, which implies the meaning and context behind the exact word match. Meanwhile, the Keyword and Concept Matching Module verifies the essential words and concepts in the reference answer based on such techniques as TF-IDF or RAKE and finds its presence in the student response. The Scoring Module combines both semantic similarity and keyword coverage in a weighted formula that marks out the marks which includes the possibility of partially correct answers as partial marks. Lastly, the Feedback and Output Module issues a report on

evaluation, which points out the key points missed and scores. It is an architecture that provides a fair, scalable, and transparent grading process that can process large amounts of handwritten answer sheets with minimal human intervention.

AI Answer Sheet Evaluator - System Architecture Flowchart

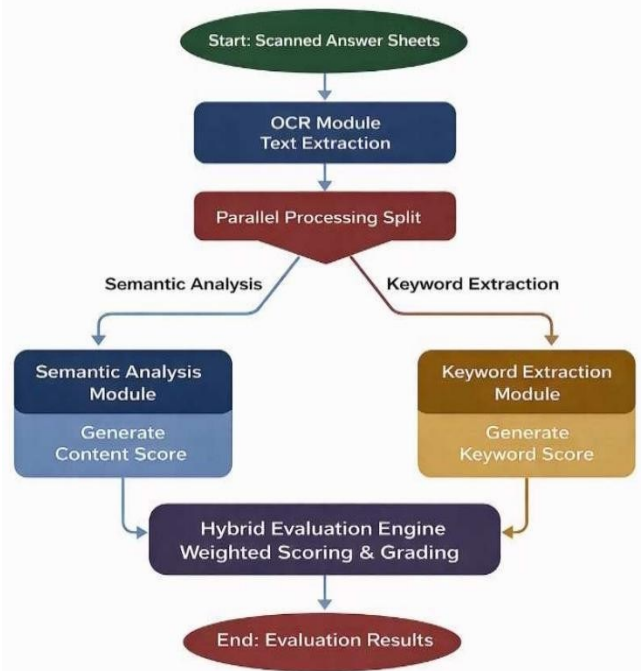


Fig. 1 System Architecture

Fig.1 referring the automation of hand-written answers evaluation by use of OCR and NLP. Tesseract and TrOCR are used to turn handwritten answers into text, preprocessed, and batch-wise semantically matched with model answers with Sentence-BERT and GPT. The assigning of marks is in accordance with the similarity, relevance, and completeness with partial grading assistance. The end results and reports are arrived at and stored safely, making the evaluation to be quick and free of bias.

C. Module Training And Data Utilization

- This module handles the login and registration of the staffs as well as the students. The members of the staff are able to log in to post questions and reference answers safely and the students log in to view questions and post their answers. Appropriate authentication protects privacy of data and only authorized users can access the data.
- Handwritten responses for evaluation. Students begin by selecting the relevant question from a dropdown menu, ensuring the answer is correctly mapped to its corresponding reference answer. They can then upload a clear image of their written response through the upload interface. Once submitted the system processes the image using OCR to extract the text and sends it to the AI evaluation pipeline for semantic analysis and scoring. This module provides a simple, user-friendly way for students to submit their answers digitally and ensures all submissions are securely stored for evaluation.
- Tesseract OCR is an OCR module that can be used to

read and convert handwritten answer images into machine-readable text. This is an important step since the text that has been identified in the picture is used as the input in semantic analysis. The preprocessing functions dealt with by the module include noise removal, and text enhancement to enhance the accuracy of OCR.

AI Answer Sheet Evaluator - System Flow

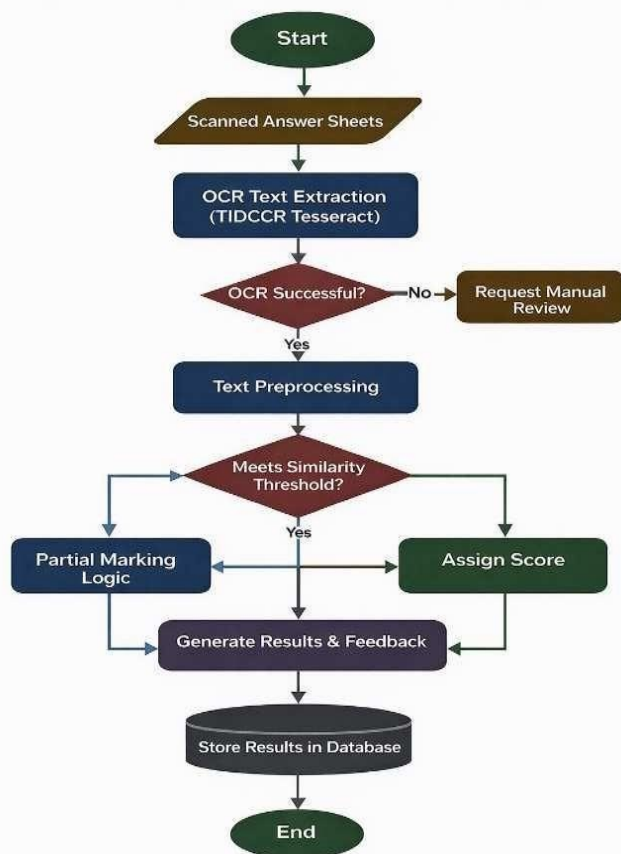


Fig. 2 Module Layers

Fig.2 is the layers and Module Interaction Diagram of the AI Answer Sheet Evaluator depicts a vertical flow of processes in terms of input and output. It begins with the Input and Scanning Module that scans the answer sheets and model answers, done by hand. The results are then passed through OCR, Preprocessing and Semantic Analysis layers of extracting and understanding text. The Evaluation Module uses the similarity and relevance to give full or partial marks. Lastly, the Result Generation and Database Layers evaluate results, create feedback and store the information to be analyzed in the future so that there is a completely automated and efficient evaluation system

D. Bert Model

BERT (Bidirectional Encoder Representations of Transformers) model is at the center of the project to evaluate, automatically, the descriptive student answers. BERT is a Google-created transformer-based NLP model, which is read in both directions, i.e., the model understands the context based on both sides of a sentence: on the left and right sides. Such two-way processing enables BERT to extract rich semantic information, thus becoming much more useful when compared to other more traditional

approaches to keyword- based search, like TF-IDF or Bag-of-Words, which can be easily subverted by students through the use of a synonym, paraphrasing, or alternative sentence structure. In this project, the reference answer submitted by the teacher and the student answer retrieved using OCR are both inputted in BERT in the form of a postprocessing and tokenizing. Extra tokens like the [CLS] and [SEP] are introduced and BERT transforms every answer into high-dimensional contextual representations.

The overall sentence representation of both answers is the token embedding of [CLS]. A measure of cosine similarity is then used to compare the two embeddings, which yields a similarity score, which indicates the extent of the similarity of the two answers between the student and the reference answer. Lastly, the similarity score is converted to marks, which guarantees equal, objective, and human evaluation. BERT therefore allows semantic comparison to be done correctly and BERT is the intelligence backbone of the answer evaluation system. Sentence-BERT (SBERT) and GPT were chosen in this project due to the balance between semantic comprehension, performance, and usefulness in performing practical assessment of descriptive student responses.

SBERT is directly created to be used in sentence-level similarity, being more appropriate in tasks of comparing student responses with model answers. SBERT generates fixed embeddings to compute similarity, unlike RoBERTa, DeBERTa, or BART, which rely on costly pairwise cross-encoding to do so, making SBERT inference significantly faster, and its cost is low enough to be run on real-time grading. GPT builds on SBERT because it provides contextual reasoning, paraphrase comprehension and linguistic fluency, which allows it to understand answers expressed incompletely or conceptually. RoBERTa and DeBERTa are not semantically similarity scoring optimized despite providing strong token-level accuracy. BART is not designed to evaluate but produces texts and reconstructs. A combination of SBERT and GPT has high semantic accuracy, strong support of paraphrased responses, stronger partial-marking, and can scale well to large batches of scripts with answers.

V. PERFORMANCE ANDEVALUATION

The results of AI Answer Sheet Evaluator are assessed with the help of various evaluation measurements which aim at the accuracy and fairness of the grading. The semantic accuracy measures how the system evaluation is similar to the answers given by the teacher by calculating the cosine similarity between embeddings of the student responses and the model answers. The coverage of keywords measures how the necessary concepts or important terms of the reference answer appear in the answer of the student so that the students are given credit of having discussed key aspects but differently put.

Partially correct answers can be scored as partial marks which is a feature of the system and does not provide a score of right or wrong, but fair evaluation of partly correct answers. Moreover, the measures of precision and recall are used to indicate the ratio of correctly identified points which are correct of all the identified points and the ratio of identified points of all the expected points respectively with the F1-score giving a single value which represents the whole performance. Processing time is also kept in check so

that the system can effectively handle large batches of answer sheets, consistency and bias check is done so that similar answers are scored in a consistent manner across several submissions. Collectively, these measures offer a multi-faceted analysis of the system, and they show it as accurate, efficient, fair, and scalable to automated analysis of handwritten answers.

A. Similarity Matrix:

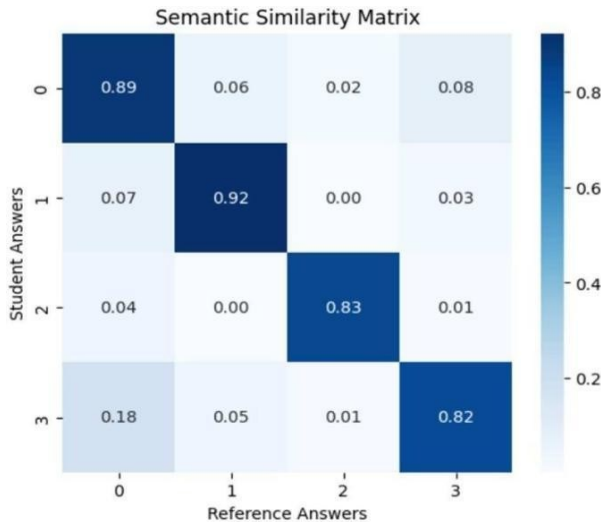


Fig. 3 Similarity Matrix

Fig.3 is the results of an evaluation of how accurately the answer of each student corresponds to the meaning of the reference answer based on Sentence-BERT embeddings and cosine similarity. It also captures the contextual/conceptual understanding and not word matching. When the values are high it means more semantic alignment. It makes sure that students with various phrasing do not get evaluated unfairly as they all serve the same purpose. It is through this matrix that the system is based on meaning-based assessment.

B. Precision Matrix

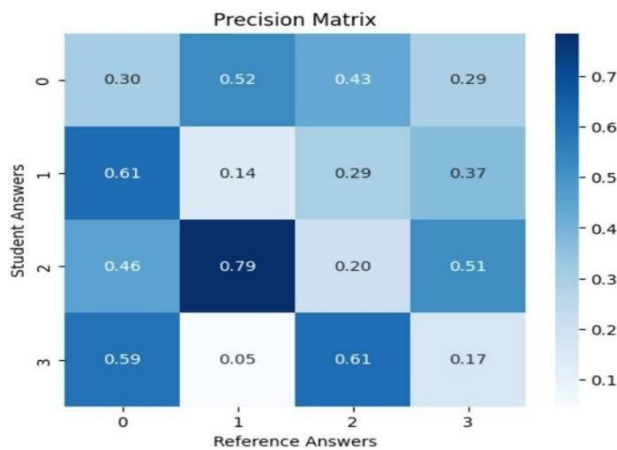


Fig. 4 Precision Matrix

Fig.4 is the proportions of the number of relevant concepts correctly identified by the system from the number of relevant concepts that the system marked as relevant. It measures the capability of the system to prevent over-marking wrong or irrelevant material. An increase in the values of higher precision means that there is a better identification of valid components of answers. This is because this matrix guarantees that only correct

information is rewarded in the model. It helps to increase the reliability of the evaluation.

VI. RESULT AND ANALYSIS

The proposed AI Answer Sheet Evaluator system will provide a quick, objective, and smart system of evaluating handwritten answer sheets. The application of OCR, NLP and semantic analysis will enable the system to effectively read the handwritten responses into digital text and evaluate them on the basis of the meaning and the coverage of the concepts other than a simple word-to-word matching. It will produce accurate score matrices of every assessment measure which are semantic similarity, key word coverage, precision, recall, F1 score and partial marking. The teachers will get very specific feedback report on how the marks were given and which aspects were not given in the process which will increase the transparency in the marking process. It is likely to greatly decrease the time of evaluation, remove human biases, and guarantee uniform scoring on high volumes of answer sheets, which is why it is very appropriate to schools, universities, and higher examination procedures.

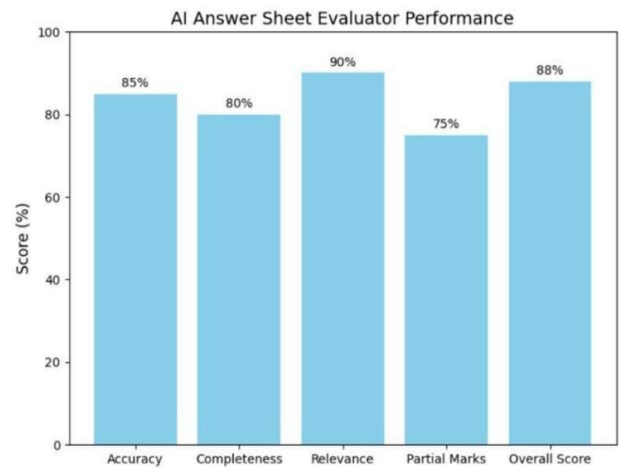


Fig. 5 Performance Graph

Fig.5 provide percentage based performance on the major evaluation measure like accuracy, relevancy, completeness, and partial marking. It emphasizes stable and efficient performance of the system, which evidently depicts the performance effectiveness and reliability of the automated grading system.

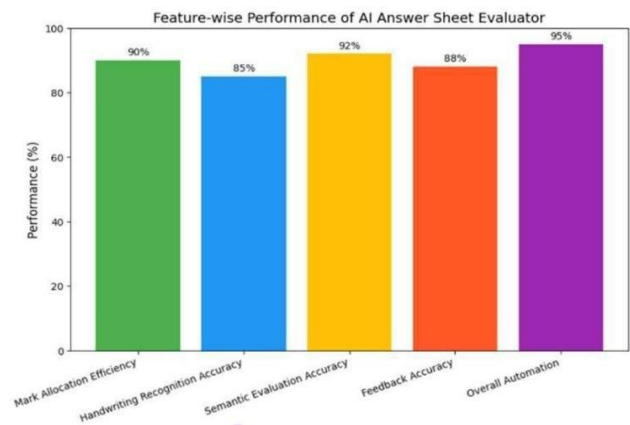


Fig. 6 Feature-wise Performance

Fig.6 is the performance graph in terms of features is used to compare the percentage efficacy of all the important system functionalities. It indicates good results in the areas of handwriting recognition, semantic assessment and automatic mark allocation, which proves that the system is efficient and reliable.

VII. CONCLUSION

The AI Answer Sheet Evaluator proves effective as it shows how it is possible to change the old-fashioned ways of academic assessment using Artificial Intelligence, OCR, and Natural Language Processing. The system can turn handwritten responses into digital text and analyze them with semantic similarity models (Sentence-BERT and GPT) instead of keywords, and all it takes is to use OCR tools (Tesseract and TrOCR) to digitize the input response. This allows grading descriptive answers fairly, consistently and transparently, unlike human assessment which is usually subjective and tired. The introduction of a partial marking system also increases the feasibility of the system as it gives the partially correct answers proportional marks to further increase fairness in the overall grading. All in all, the project also provides scalability, whereby institutions can handle high volumes of answer scripts effectively, evaluator bias is removed, and reliable and automated assessment is conducted. The system is the basis of AI-driven academic assessment tools in the future, as well as potentially being used on a large scale in schools, universities, and high-stakes testing facilities.

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